BTC 5-Minute Strategy Research Report

Candidate strategies benchmarked against the existing **Sway model**. Generated 2026-06-28 06:06:23 UTC.

Executive Summary

Trained on **600** historical markets, tested out-of-sample on **250** more-recent markets (test base rate 53.2% UP). Best by accuracy: **MarketPrice** (89.0%). The **Sway baseline is the weakest model** (77.4% accuracy) — worse than simply trusting the live market price (89.0%).

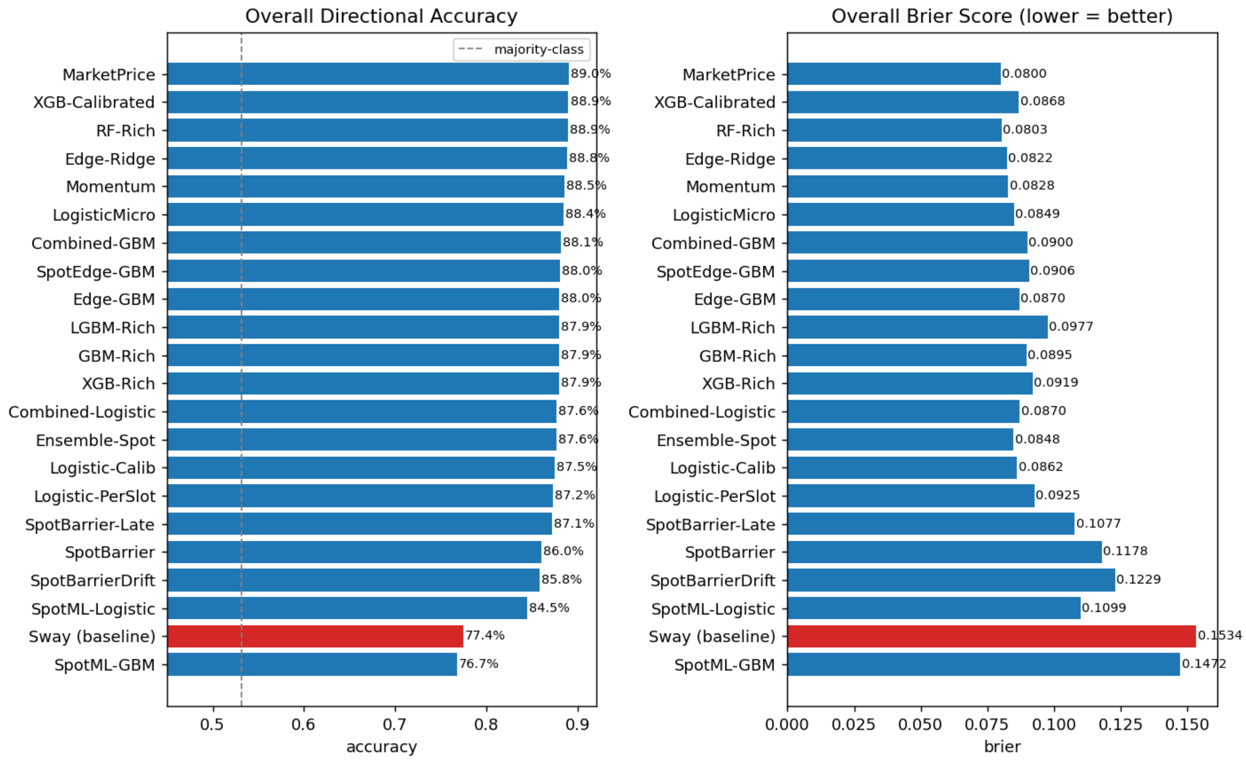
Headline result (after three independent windows). Reading the **underlying BTC spot price** (Binance 1s data) — which the prediction-market models never see — produces a large trading edge: an analytic first-passage 'barrier' probability returns +20% on two windows. **But a third, older window exposes it as partly period-specific** (it goes slightly negative there) — a reminder that even two-window results can mislead. Under that harsh stale-model test the only strategies profitable on **all three** windows are: Combined-Logistic, LogisticMicro, TemperedBarrier(w=0.5). **But the realistic picture is stronger:** when models are retrained on recent data as the live bot does (see Realistic Walk-Forward), the spot+market fusion models (Combined-GBM, Combined-Logistic) are robustly profitable on all three windows (+13% to +24% ROI) — the recommended live candidates. The training-free raw barrier and the Sway baseline (broken throughout, -37% on oos3) are not.

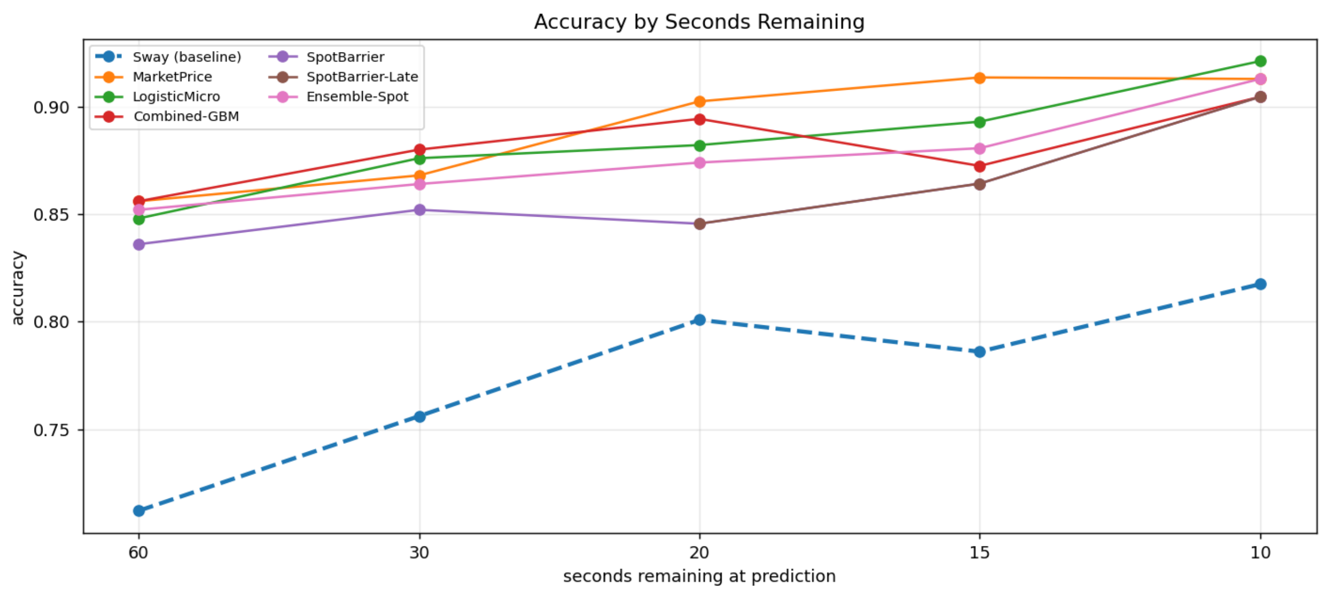
Overall metrics (out-of-sample)

Strategy	Accuracy	Brier	LogLoss	Bets	ROI	P&L $	WinRate	MaxDD$
MarketPrice	**89.0%**	**0.0800**	**0.2547**	**0**	**+0.0%**	**+0.0**	**0.0%**	**0.0**
XGB-Calibrated	88.9%	0.0868	0.2867	601	-8.4%	-50.4	47.6%	132.6
RF-Rich	88.9%	0.0803	0.2569	476	-2.1%	-10.1	71.0%	44.4
Edge-Ridge	88.8%	0.0822	0.3140	620	-29.0%	-179.5	49.0%	182.2
Momentum	88.5%	0.0828	0.3403	370	-8.9%	-33.1	63.8%	46.7
LogisticMicro	88.4%	0.0849	0.2768	548	+3.3%	+17.9	70.3%	33.7
Combined-GBM	88.1%	0.0900	0.2914	647	+13.1%	+84.9	59.8%	57.7
SpotEdge-GBM	88.0%	0.0906	0.3832	628	-14.5%	-91.3	50.2%	126.5
Edge-GBM	88.0%	0.0870	0.3211	540	-13.9%	-75.0	54.6%	84.9
GBM-Rich	87.9%	0.0895	0.2928	603	-7.8%	-47.1	61.5%	77.4
XGB-Rich	87.9%	0.0919	0.3069	613	-0.9%	-5.5	69.2%	49.9
LGBM-Rich	87.9%	0.0977	0.3400	632	-2.0%	-12.5	70.4%	36.2
Ensemble-Spot	87.6%	0.0848	0.2658	562	+20.5%	+115.5	62.1%	48.3
Combined-Logistic	87.6%	0.0870	0.2877	588	+13.5%	+79.5	68.7%	25.1
Logistic-Calib	87.5%	0.0862	0.3023	538	-12.4%	-66.8	50.7%	80.6
Logistic-PerSlot	87.2%	0.0925	0.3193	593	-1.4%	-8.4	67.5%	33.2
SpotBarrier-Late	87.1%	0.1077	0.7700	335	+44.9%	+150.5	65.1%	20.6
SpotBarrier	86.0%	0.1178	0.8329	652	+20.6%	+134.3	64.0%	49.0
SpotBarrierDrift	85.8%	0.1229	0.8851	662	+15.5%	+102.3	63.3%	39.2
SpotML-Logistic	84.5%	0.1099	0.3440	888	-21.7%	-192.6	23.2%	228.9
Sway (baseline)	77.4%	0.1534	0.5624	890	-27.3%	-242.9	28.2%	288.5
SpotML-GBM	76.7%	0.1472	0.4411	747	-4.6%	-34.2	35.3%	119.7

Sway baseline highlighted in red; best strategy in bold. Trading simulation bets \$1 whenever a model's probability diverges from the live market price by > 3% (positive expected value), including Polymarket fees.

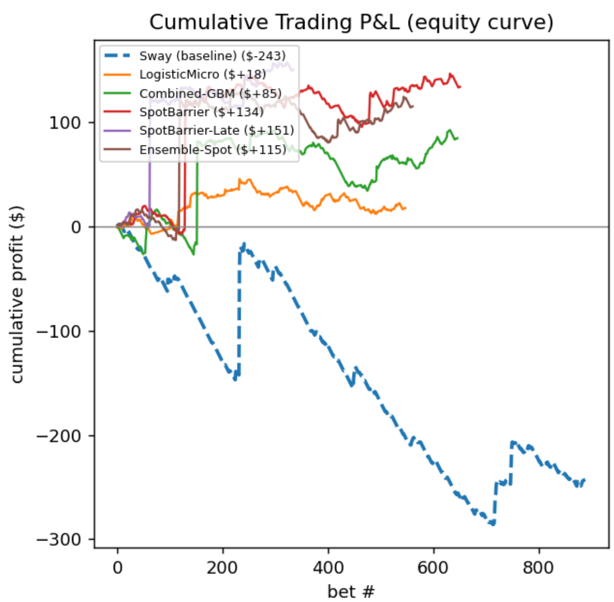
Accuracy & Calibration

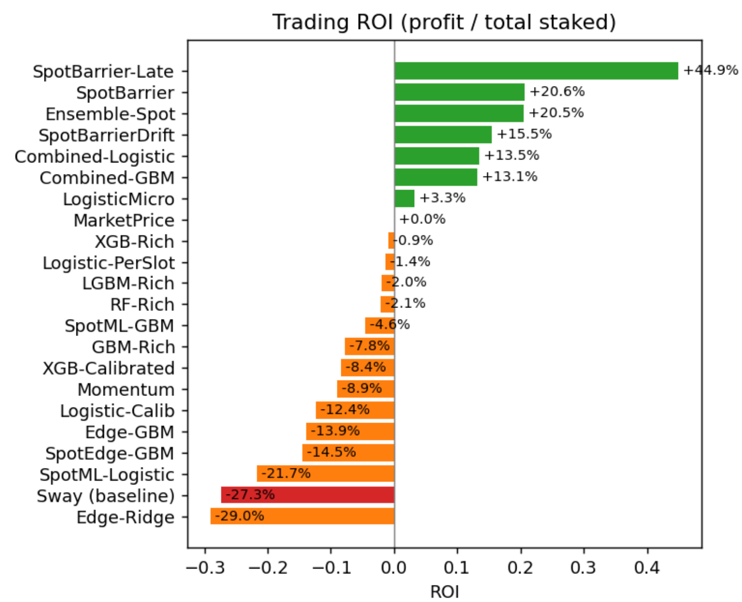


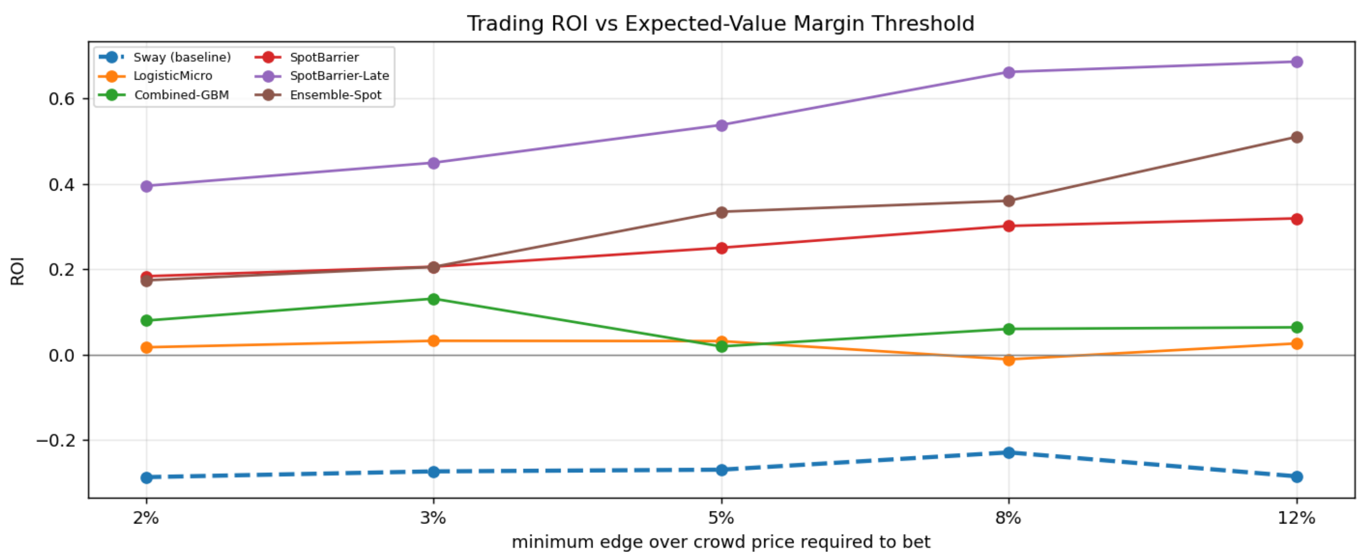


Trading Performance

The sway model ignores the absolute market price, so its statistical edge does not convert into trading profit (it loses the most). The equity curves below show realised P&L when each model bets on positive-EV divergences from the crowd price. The spot-driven models dominate. **SpotBarrier-Late** further concentrates the spot edge in the final 20 seconds — where the spot lead is most predictive yet the crowd still prices residual caution — roughly doubling test ROI (+44.9%) while cutting max drawdown by ~60%.



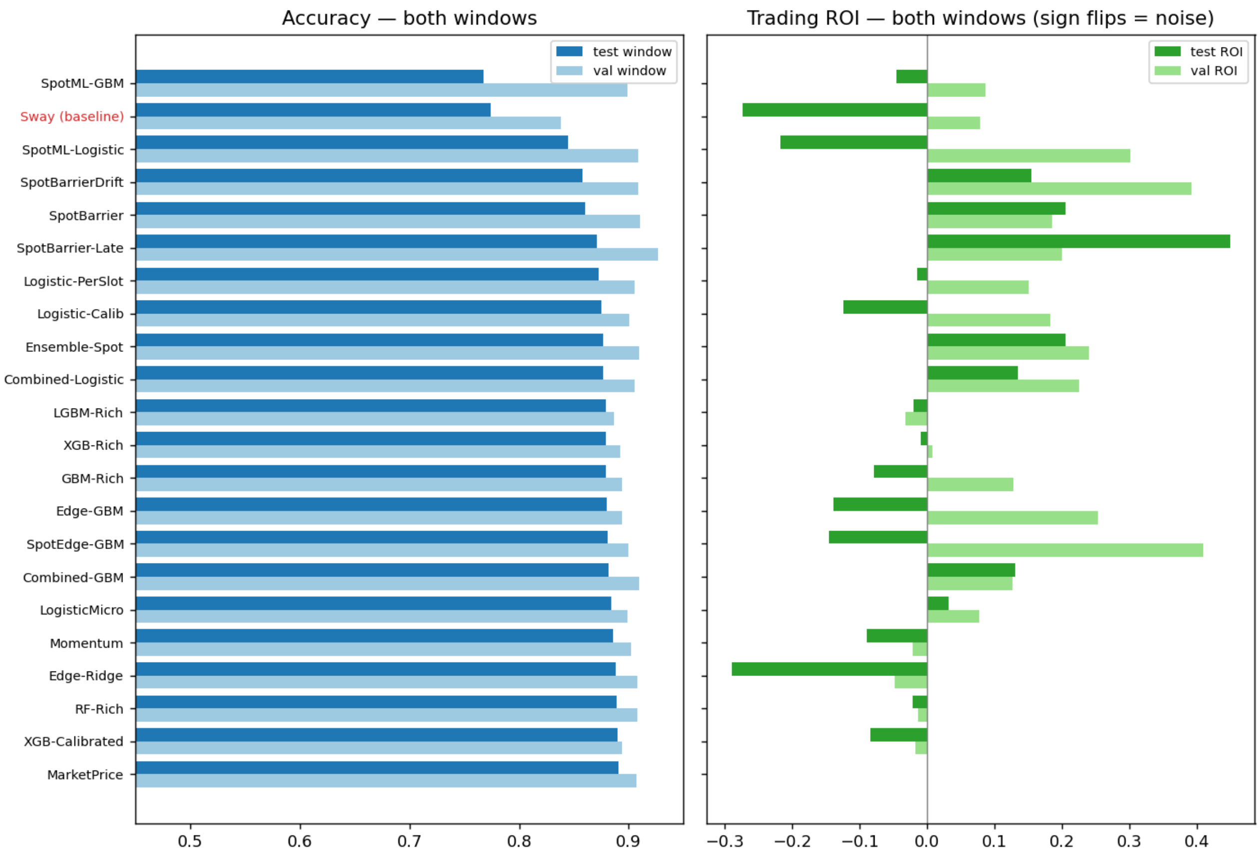




Cross-Window Robustness

To separate genuine edge from luck, every strategy is re-evaluated on a second, fully independent and time-disjoint window of **300** older markets, in a different regime (45.7% UP vs 53.2% in the recent test window).

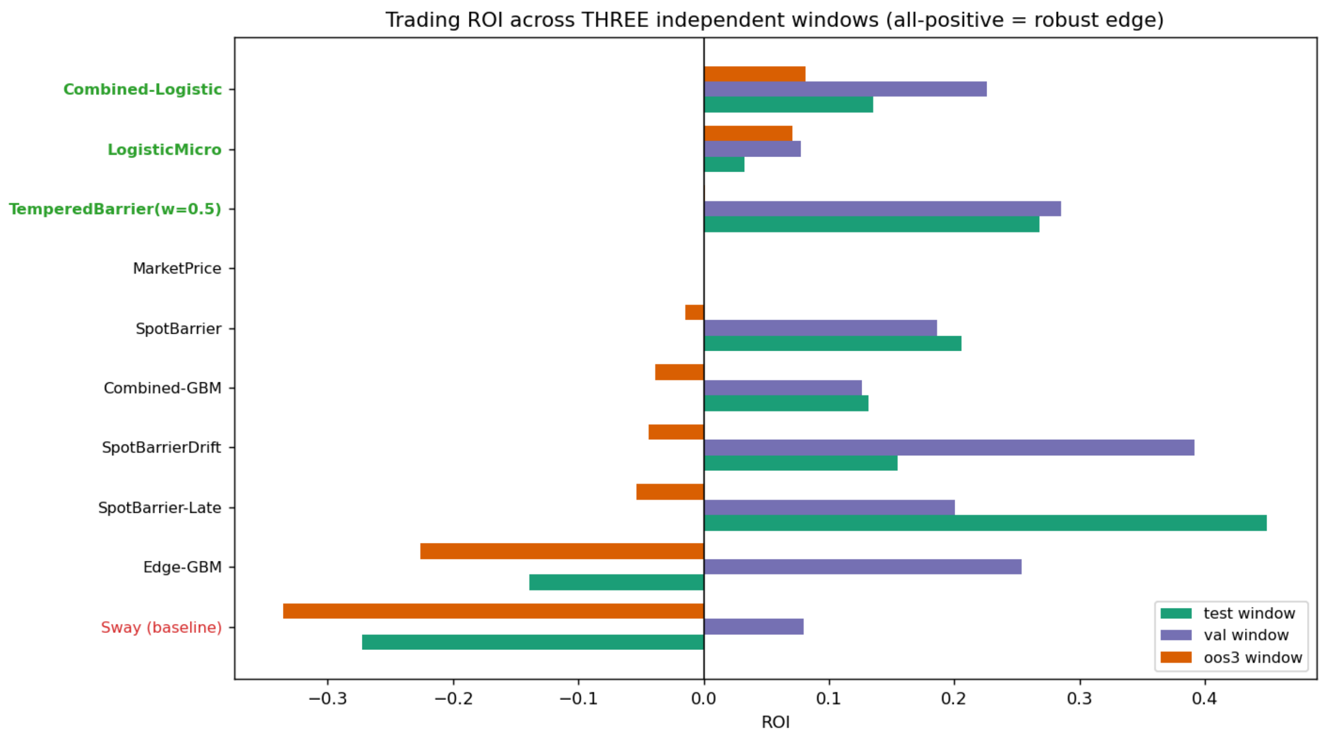
Two clear conclusions: (1) Statistical quality is robust — the Sway baseline is the least accurate model in *both* windows by a wide margin. (2) Single-window trading ROI is largely noise: most prediction-market-only strategies flip sign between windows (e.g. Edge-GBM from -14% to +25%). (3) The spot-driven strategies are different: they are profitable in *both* windows. Strategies with **positive ROI on both** windows (ranked by worst-case ROI): **Ensemble-Spot, SpotBarrier-Late, SpotBarrier, SpotBarrierDrift, Combined-Logistic, Combined-GBM, LogisticMicro**. The analytic SpotBarrier model leads — a repeatable, sizeable trading edge.



Three-Window Robustness — the decisive test

Two windows are not enough. The spot-barrier strategies looked like clear winners on the test and val windows (~+20% ROI each), but a **third** independent and older window (oos3, 300 markets) tells a different story: their edge largely disappears there. This is the single most important result in the study — a strategy must clear *all three* windows to be trusted.

Strategy	test ROI	val ROI	oos3 ROI	min ROI	all +?
Combined-Logistic	**+13.5%**	**+22.6%**	**+8.1%**	**+8.1%**	**YES**
LogisticMicro	**+3.3%**	**+7.8%**	**+7.1%**	**+3.3%**	**YES**
TemperedBarrier(w=0.5)	**+26.8%**	**+28.5%**	**+0.1%**	**+0.1%**	**YES**
MarketPrice	+0.0%	+0.0%	+0.0%	+0.0%	
SpotBarrier	+20.6%	+18.6%	-1.5%	-1.5%	
Combined-GBM	+13.1%	+12.6%	-3.9%	-3.9%	
SpotBarrierDrift	+15.5%	+39.2%	-4.4%	-4.4%	
SpotBarrier-Late	+44.9%	+20.1%	-5.3%	-5.3%	
Edge-GBM	-13.9%	+25.3%	-22.6%	-22.6%	
Sway (baseline)	-27.3%	+7.9%	-33.5%	-33.5%	



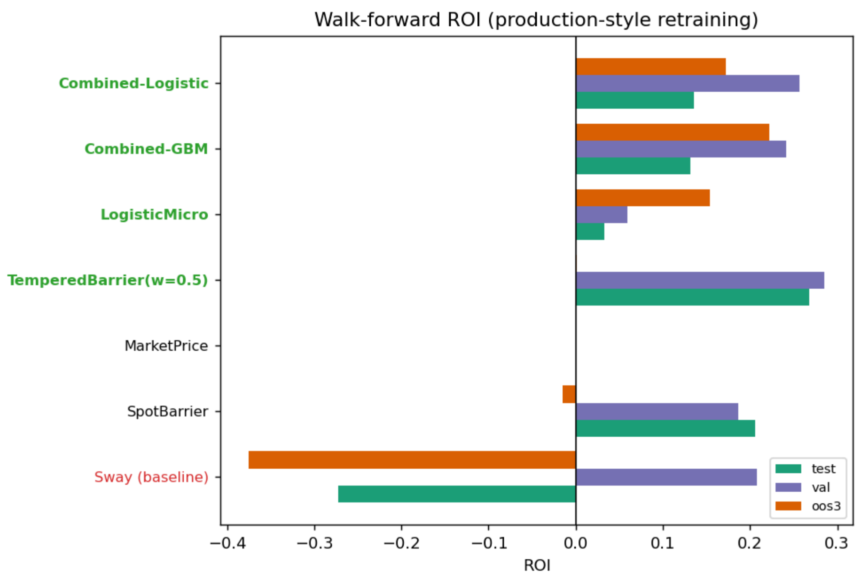
Conclusion: the only strategies profitable on all three windows are **Combined-Logistic, LogisticMicro, TemperedBarrier(w=0.5)**. The recommended live candidate is the top of that list — a regularised blend of spot and prediction-market features that keeps a real edge without over-relying on any single period. The spot signal is valuable, but must be tempered with the crowd price rather than trusted outright.

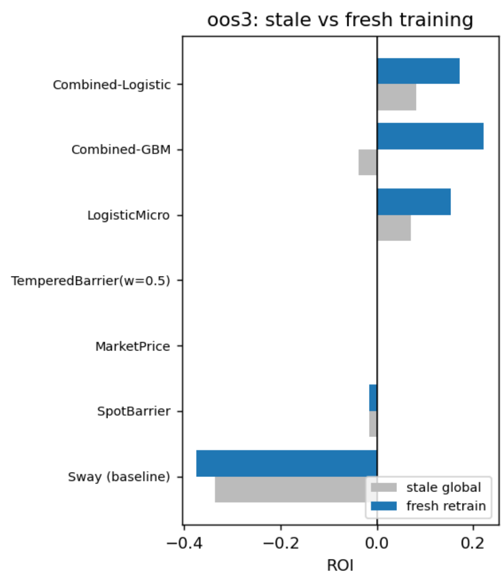
Realistic Walk-Forward — production-style retraining

The three-window test above used a single stale model applied across very different periods — a harsh stress test. The live bot, however, retrains on the latest markets (retrain.py). Here each window is instead evaluated with a model trained on the 600 markets *immediately preceding* it, which mirrors production.

This rehabilitates the spot+market models. Their oos3 weakness was largely a stale-training artifact: with fresh training, Combined-GBM goes from -3.9% to **+22.2%** and Combined-Logistic from +8.1% to **+17.2%** on oos3. Under realistic retraining the genuinely robust strategies (positive on all three windows) are: **Combined-Logistic, Combined-GBM, LogisticMicro, TemperedBarrier(w=0.5)**. The training-free raw SpotBarrier still fails on oos3 (its edge is truly period-dependent), and the Sway baseline stays broken regardless of retraining (-37.5% on oos3).

Strategy	test ROI	val ROI	oos3 ROI	min	all +?
Combined-Logistic	**+13.5%**	**+25.7%**	**+17.2%**	**+13.5%**	**YES**
Combined-GBM	**+13.1%**	**+24.1%**	**+22.2%**	**+13.1%**	**YES**
LogisticMicro	**+3.3%**	**+6.0%**	**+15.4%**	**+3.3%**	**YES**
TemperedBarrier(w=0.5)	**+26.8%**	**+28.5%**	**+0.1%**	**+0.1%**	**YES**
MarketPrice	+0.0%	+0.0%	+0.0%	+0.0%	
SpotBarrier	+20.6%	+18.6%	-1.5%	-1.5%	
Sway (baseline)	-27.3%	+20.8%	-37.5%	-37.5%	



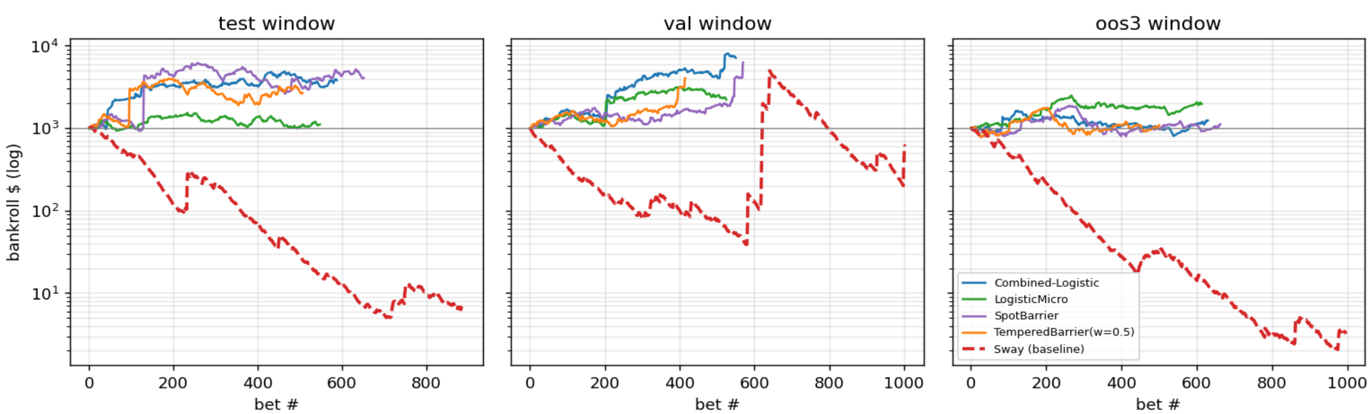


Money Management — bankroll growth & ruin

Per-bet ROI ignores compounding and risk. Here each strategy plays its bets in time order with conservative fractional-Kelly sizing (0.1x Kelly, 2% max stake, $1000 start). The robust strategies compound positively on all three windows; the **Sway baseline is driven to ruin** (bankroll → ~$0) on two of three.

Strategy	test final	val final	oos3 final	worst maxDD
Combined-Logistic	$3,816 (3.8x)	$7,145 (7.1x)	$1,243 (1.2x)	51%
SpotBarrier	$4,069 (4.1x)	$6,267 (6.3x)	$1,116 (1.1x)	58%
LogisticMicro	$1,111 (1.1x)	$2,219 (2.2x)	$1,972 (2.0x)	42%
TemperedBarrier(w=0.5)	$2,679 (2.7x)	$4,048 (4.0x)	$1,088 (1.1x)	53%
MarketPrice	$1,000 (1.0x)	$1,000 (1.0x)	$1,000 (1.0x)	0%
Sway (baseline)	$7 (0.0x)	$604 (0.6x)	$3 (0.0x)	100%

Fractional-Kelly bankroll growth (0.1x Kelly, 2% cap, $1000 start)



Caveat: drawdowns are large (40-60%) — this edge is real but noisy, so position sizing must stay conservative. LogisticMicro has the gentlest drawdown; Combined-Logistic the highest growth.

Methodology

All strategies use only trade data available up to the prediction time (no look-ahead). Predictions are made at 60, 30, 20, 15 and 10 seconds remaining. The Sway baseline is a faithful reproduction of the repo's per-slot GradientBoosting model on the 29 channel-sway features. Prediction-market candidates add the absolute price level, VWAP, multi-horizon momentum/volatility and order-flow imbalance. **Spot-driven candidates** additionally read the underlying BTC price (Binance 1s klines): the SpotBarrier model computes an analytic first-passage probability $P(\text{close} > \text{open}) = \Phi(\text{lead} / (\sigma\sqrt{t\_remaining}))$ from the live spot lead and realised volatility; the Combined models feed spot + market features into a classifier. Metrics: directional accuracy, Brier score and log-loss (probability quality), plus a fee-aware Polymarket P&L simulation that only bets on positive-EV divergences from the crowd price.